MD Shihabul Islam Shovo

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Profile

Computer Science and Engineering graduate and research assistant specialising in **lightweight deep learning for medical imaging and biosignal analysis**. Author of **3 accepted**, **2 under-review**, and **2 in-preparation** peer-reviewed papers spanning dermatology, ophthalmology, radiology, EEG, and wearable IMU sensing, with recurring contributions in parameter-efficient architecture design, structural reparameterization, semi-supervised learning, statistical validation, and explainable AI (Grad-CAM). Complements this with hands-on full-stack engineering experience (Django, React, REST APIs) for building the tooling and pipelines that research prototypes need.

Research Interests

Medical Image Analysis · Lightweight & Efficient Deep Learning · Semi-Supervised Learning · State-Space Models (Mamba) · Biosignal & Wearable Sensing · Explainable AI

Education

B.Sc. in Computer Science and Engineering — *2020 – 2024*
Ahsanullah University of Science and Technology (AUST) — *Dhaka, Bangladesh*
CGPA: **3.078 / 4.00**

Publications

Accepted (3) — *peer-reviewed and accepted for presentation at IEEE conferences*

[C1] N. K. Paul, **M. S. I. Shovo**, I. J. Esha, and A. Rahman, "SleepEffFormer: Efficient CNN-Transformer with Transition-Aware Smoothing for Single-Channel EEG Sleep Stage Classification," *5th IEEE Int. Conf. on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, Sept. 2026. *Result:* 83.9% accuracy, 78.9% macro-F1, $\kappa = 0.765$ on Sleep-EDF Expanded (78 recordings); matches AttnSleep with 3–5× fewer parameters (∼367K).

[C2] **M. S. I. Shovo**, N. K. Paul, A. Rahman, and I. J. Esha, "WaveFoG: Wavelet-Gated Transformer for Parkinson's Freezing of Gait Detection Using Wearable Accelerometer Signals," *5th IEEE BECITHCON*, Dhaka, Bangladesh, Sept. 2026. *(First author) Result:* F1 0.875 ± 0.017, AUPRC 0.833 ± 0.020 on the Kaggle TLVMC FoG dataset; up to +5.4pp F1 over CNN/BiLSTM/Transformer baselines under subject-independent 10-fold CV.

[C3] NIMISHES Lab, "DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion," *1st Int. Conf. on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONEECT)*, 2026. *Result:* $88.77 \pm 0.61\%$ accuracy, QWK 0.884 on APTOS 2019 with only 4.69M parameters (5× smaller than ResNet50); Grad-CAM explainability.

Under Review (2) — *submitted to the 29th IEEE Int. Conf. on Computer and Information Technology (ICCIT 2026)*

[U1] NIMISHES Lab, "AWEF-Net: An Attention-Weighted Ensemble Fusion Framework for Multi-Class Pulmonary Disease Classification from Chest Radiographs," *29th IEEE ICCIT*, 2026, under review. *Result:* 96.3% macro-F1, 97.1% COVID-19 recall on 21,165-image CXR dataset; adaptive attention gate adds under 0.6% of total model parameters.

[U2] NIMISHES Lab, "PDGuard: A CNN-BiLSTM Attention Network for Wearable IMU-Based Parkinson's Disease Detection," *29th IEEE ICCIT*, 2026, under review. *Result:* $88.34 \pm 1.01\%$ accuracy, ROC-AUC 0.918 on the PADS cohort (469 participants); 1.24M parameters, 18.3 ms/window on CPU.

In Preparation (2) — *targeting submission in late 2026*

[P1] **M. S. I. Shovo**, N. K. Paul, and K. Morol, "LCM-UNet: A Reparameterizable Local-Compensated Mamba U-Net for Skin Lesion Segmentation," journal manuscript, target submission Sept. 2026. *(First author) Result:* 34,180 parameters, 0.0597 GFLOPs after reparameterization; +0.48/+0.37pp IoU over a retrained baseline on ISIC 2017/2018; external PH2 transfer $86.37 \pm 0.29\%$ IoU; Wilcoxon significant at $p < 10^{-4}$.

[P2] N. K. Paul, **M. S. I. Shovo**, and C. Gonzalez, "CoMAF-Polyp: Calibrated Reliability-Aware Fusion of Specialist and Foundation Pseudo-Labels for Semi-Supervised Polyp Segmentation," target: *WACV 2027 Workshop (Computer Vision Foundation)*. *Result:* 88.46% Dice on Kvasir-SEG with 10% labels, recovering 98.6% of a per-image oracle; +2.44pp average Dice over the best baseline across four unseen datasets under domain shift.

Research Experience

ELITE Research Lab LLC — *Jun 2026 – Present*
Junior Research Assistant (Aug 2026 – Present); Research Intern (Jun 2026 – Aug 2026) — *Queens, NY, USA (Remote) · Part-time*

Project: LCM-UNet – Lightweight Skin Lesion Segmentation using a Mamba-based U-Net

- Designed and led development of an ultra-lightweight (∼34K-parameter, 0.06 GFLOPs) Mamba-based state-space U-Net for melanoma-related lesion boundary detection, targeting deployment on low-cost point-of-care hardware.
- Proposed the Residual Dilated Local Adapter (RDLA), a zero-initialized dilated convolution module that adds optimisation capacity during training and is exactly fused back into the base kernel via structural reparameterization before inference, at zero added deployment cost.
- Validated the approach with multi-seed statistical testing (image-level Wilcoxon, $p < 10^{-4}$) on ISIC 2017, ISIC 2018, and external PH2 datasets, demonstrating consistent boundary-quality improvements over the baseline.

- Promoted from Research Intern to Junior Research Assistant in recognition of research contributions.

NIMISHES Lab — *Lead Researcher* *2026 – Present*

- Contributed to 6 peer-reviewed manuscripts across medical imaging (dermatology, ophthalmology, radiology) and wearable biosignal analysis (EEG, IMU) as part of an independent research lab building explainable, deployment-efficient AI for clinical use.
- Contributed to model design, cross-validation and statistical significance testing, and Grad-CAM-based explainability analysis across projects, consistently achieving state-of-the-art parameter efficiency versus standard backbones (e.g. ResNet50, DenseNet201).

Professional Experience

Web & Automation Engineer — *Cleaning Connected* *Jan 2025 – Present*
California, USA
(Remote) · Part-time

- Designed and developed digital solutions supporting business operations, including website development, lead management automation, and workflow optimisation.
- Maintained web platforms, integrated business tools, and resolved technical issues to improve operational efficiency.

Technical Skills

AI/ML & Research	PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Mamba / state-space models, Grad-CAM/XAI, statistical validation (Wilcoxon), structural reparameterization
Languages	Python, JavaScript, C#, Java, C/C++
Frontend	React, Tailwind CSS, Bootstrap, TypeScript, HTML, CSS
Backend	Django, Django REST Framework, ASP.NET MVC, Flask, PHP
Databases	SQLite, MySQL, PostgreSQL, Microsoft SQL Server
Tools & Auth	Git, GitHub, Postman, Selenium WebDriver, JWT Authentication

Selected Software Development Projects

Hospital Management System — *DRF, React, JWT, SQLite/PostgreSQL* GitHub

- Designed and built a role-based hospital management platform (admin, doctor, patient, receptionist) with a Django REST Framework API and a React (Vite) SPA, modeling 9 relational entities including patients, doctors, appointments, prescriptions, and billing.
- Implemented JWT authentication with automatic silent token refresh (with request deduplication) and dual-layer role-based authorization — queryset-level scoping plus object-level permission checks — ensuring users can only access their own records.
- Built a consistent API error-handling layer (custom DRF exception handler) with auto-generated OpenAPI/Swagger docs via drf-spectacular, reusable React data-fetching hooks, and a centralized axios client with request/response interceptors.

MRI Tumor Detection System — *Flask, TensorFlow, Keras, VGG16* GitHub

- Web-based AI system classifying MRI brain scans into four tumor categories, with confidence scoring and automated PDF report generation.

Interactive Teaching Platform — *Django, DRF, Tailwind CSS* GitHub

- Interactive learning platform with clickable terms, multimedia popups, REST APIs for dynamic content, and multilingual SEO-friendly URL slugs.

Additional projects: EasyHome, house rental management system (ASP.NET MVC, C#, MSSQL) — NOTES App (Django REST Framework, React, JWT) — Exclusive Shop, e-commerce frontend (React, TypeScript) — Lift, e-commerce platform (PHP, MySQL) —

Courses & Professional Training

Servercamp — *Astha IT* *May 2026 – Ongoing*
Focus: Backend development (4-month program) with C#, OOP, ASP.NET, and SQL.

Full Stack Web Development with Python, Django & React *Oct 2025 – Ongoing*
Focus: JavaScript, Python, HTML, CSS, Django, and React.

Industrial Robot Operation and Troubleshooting — *BITAC, Dhaka* *Jul 2025 – Oct 2025*
Focus: Robot simulation, robot programming, troubleshooting, and Malfabasic language. *Tejgaon, Dhaka*

Competitive Programming & Extracurriculars

Codeforces · LeetCode

- Participated in Intra AUST Programming Contest, Fall 2021; Campus Ambassador, GUVI Geek Networks (IITM Research Park), 2021–2023 — organized a seminar on the importance of Data Science.